La Scuola d'Italia / Huson / Physics Nickname:
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1.1 Classwork: Measurement, Precision, and Significant Figures

Reference (provided). 1 in. = 2.54 cm   1 m ≈ 3.28 ft   1 h = 3600 s
1 km = 1000 m   1 m = 100 cm   1 cm = 10 mm   1 kg = 1000 g

1. Write down the number of significant digits in each measurement.

   (a) 42   (c) 0.0058   (e) 250.0

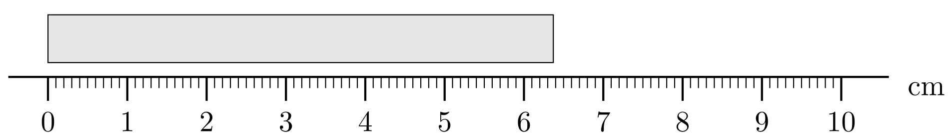
   (b) 3.60   (d) 0.01070   (f) 6.022

2. Round each value to three significant figures. Copy the calculator display followed by three dots, then round:

$$\sqrt{2} = 1.4142135\ldots \approx 1.41$$

   (a) The population of Milan: 1,371,498   (b) $\sqrt{5}$

3. The bar below is measured with a centimeter ruler marked in millimeters.



   (a) Record the length of the bar. Include one estimated digit — the last digit, the one you are not certain of.

   (b) How many significant figures does your measurement have?

   (c) Write the same length in millimeters, and again in meters.

4. Write in scientific notation, rounded to three significant figures.

   (a) The distance from New York to Rome: 6,890,000 meters.

   (b) The diameter of a red blood cell: 0.0000078 meters.

5. Write each value as an ordinary decimal number.

   (a) 4.05×10^5 grams

   (b) 2.9×10^{-4} seconds

6. Convert between units, rounding to three significant figures. Write the conversion as a fraction and show the units cancelling.

   (a) The Colosseum stands 48.5 meters tall. What is its height in feet?

   (b) A 12 inch ruler is how many centimeters long?

7. A bus on Via del Corso travels at 30 kilometers per hour. What is this speed in meters per second?

8. The mass of the Earth is 5.972×10^{24} kg and the mass of Jupiter is 1.898×10^{27} kg. How many times more massive is Jupiter than the Earth? Round to three significant figures.

9. Challenge: Estimate the number of seconds you have been alive. Give your answer in scientific notation, and explain why it would be wrong to report this number to nine significant figures.